

A negative answer to the continuous same-sign Gabor ASF problem

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Source problem

The source paper is K. Mahesh Krishna, *The (abc, pqr) -problem for Approximate Schauder Frames for Banach Spaces*, arXiv:2201.04421. At the end of Section 2 it asks:

Let $1 < p < \infty$ and q be the conjugate index of p . Characterize functions $g \in L^p(\mathbb{R})$, $h \in L^q(\mathbb{R})$ such that the pair

$$(\{\tilde{E}_b \tilde{T}_a h\}_{a,b \in \mathbb{R}}, \{E_b T_a g\}_{a,b \in \mathbb{R}})$$

is a continuous approximate Schauder frame for $L^p(\mathbb{R})$.

The same paper defines the dual pairing by

$$[x, \omega] = \int_{\mathbb{R}} x(t) \omega(t) dt$$

and defines

$$(E_b f)(t) = e^{2\pi i b t} f(t), \quad (T_a f)(t) = f(t - a),$$

with the analogous same-sign definitions for $\tilde{E}_b, \tilde{T}_a$ on L^q . The continuous ASF definition used by the source paper is the one in Krishna's arXiv:2201.00980: for every $x \in X$ and $\phi \in X^*$, the coefficient product is measurable and integrable, and the Pettis frame operator

$$Sx = \int_{\Omega} f_{\alpha}(x) \tau_{\alpha} d\mu(\alpha)$$

is bounded and invertible.

Result

Theorem 1. *Let $1 < p < \infty$, let q be conjugate to p , and let $g \in L^p(\mathbb{R})$, $h \in L^q(\mathbb{R})$. With the same-sign translation-modulation convention above, the pair*

$$(\{\tilde{E}_b \tilde{T}_a h\}_{a,b \in \mathbb{R}}, \{E_b T_a g\}_{a,b \in \mathbb{R}})$$

is not a continuous approximate Schauder frame for $L^p(\mathbb{R})$.

Proof Intuition

For Hilbert-space continuous Gabor frames, the analysis coefficient carries the opposite phase from the synthesis vector. In the source problem, both phases have the same sign with respect to the L^p - L^q bilinear pairing. When one integrates over the full frequency line, this does not produce $\delta(t-s)$; it produces $\delta(t+s)$. Thus the putative frame operator is forced to be reflection, $x(s) \mapsto x(-s)$, followed by multiplication by a correlation of h and g . That correlation is a C_0 -function, so it decays at infinity and cannot be bounded away from zero. Hence the operator cannot be invertible on $L^p(\mathbb{R})$.

Proof

Assume, toward a contradiction, that the displayed pair is a continuous ASF. Thus for all $x \in L^p$ and all $y \in L^q$, the scalar function

$$(a, b) \mapsto \left(\int_{\mathbb{R}} x(t) e^{2\pi i b t} h(t-a) dt \right) \left(\int_{\mathbb{R}} e^{2\pi i b s} g(s-a) y(s) ds \right)$$

is integrable over $\mathbb{R}^2$, and the Pettis integral defines a bounded invertible operator $S : L^p \rightarrow L^p$.

Fix $x, y \in C_c^\infty(\mathbb{R})$. For fixed a , set

$$u_a(t) = x(t)h(t-a), \quad v_a(t) = y(t)g(t-a).$$

Then $u_a, v_a \in L^1(\mathbb{R})$. With the Fourier transform convention $\widehat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi i t \xi} dt$, the two coefficients are $\widehat{u}_a(-b)$ and $\widehat{v}_a(-b)$. By the continuous ASF integrability hypothesis and Fubini's theorem, for almost every a the product $\widehat{u}_a(-b)\widehat{v}_a(-b)$ is integrable in b . For L^1 -functions whose Fourier-transform product is integrable, Fourier inversion for the product gives

$$\int_{\mathbb{R}} \widehat{u}_a(-b)\widehat{v}_a(-b) db = \int_{\mathbb{R}} u_a(t)v_a(-t) dt.$$

Therefore

$$\begin{aligned} \langle Sx, y \rangle &= \int_{\mathbb{R}} \int_{\mathbb{R}} x(t)y(-t)h(t-a)g(-t-a) dt da \\ &= \int_{\mathbb{R}} x(t)y(-t)c(t) dt, \end{aligned}$$

where

$$c(t) = \int_{\mathbb{R}} h(t-a)g(-t-a) da = \int_{\mathbb{R}} h(u)g(u-2t) du.$$

The interchange of the a - and t -integrals is justified by

$$\int_{\mathbb{R}} |h(t-a)g(-t-a)| da \leq \|h\|_q \|g\|_p$$

and the compact support of $x(t)y(-t)$.

Thus, for every $x, y \in C_c^\infty(\mathbb{R})$,

$$\langle Sx, y \rangle = \int_{\mathbb{R}} c(-s)x(-s)y(s) ds.$$

Since C_c^∞ is dense in L^q , it follows that

$$(Sx)(s) = m(s)x(-s), \quad m(s) = c(-s) = \int_{\mathbb{R}} h(u)g(u+2s) du$$

for every $x \in C_c^\infty(\mathbb{R})$.

Now $m \in C_0(\mathbb{R})$. Indeed, if $\check{g}(r) = g(-r)$, then

$$c(t) = (h * \check{g})(2t),$$

and the standard $L^q * L^p \subset C_0$ convolution fact applies for conjugate p, q with $1 < p < \infty$. In particular m is bounded and tends to zero at infinity.

The operator $x \mapsto m(\cdot)x(-\cdot)$ is bounded on L^p . Since it agrees with S on the dense subspace C_c^∞ , the boundedness of S forces

$$Sx = m(\cdot)x(-\cdot) \quad (x \in L^p(\mathbb{R})).$$

But no such operator is bounded below when $m \in C_0(\mathbb{R})$. Given $\varepsilon > 0$, choose a measurable set E of positive finite measure outside a compact set on which $|m| < \varepsilon$, and take $x = \mathbf{1}_{-E}$. Then

$$\|Sx\|_p = \|m \mathbf{1}_E\|_p \leq \varepsilon \|\mathbf{1}_E\|_p = \varepsilon \|x\|_p.$$

As ε is arbitrary, S is not bounded below, contradicting invertibility. This proves the theorem.

Verification and Limitations

The argument proves a negative answer only for the continuous full-plane problem as written in arXiv:2201.04421, using the source paper's same-sign definition of $\tilde{E}_b$ on the dual side. It does not address the discrete (abc, pqr) -problem, the discrete (MN, PQ) -problem, or a modified continuous Gabor problem with the opposite phase in the dual coefficient.

The proof is non-computational. The only analytic inputs are the continuous ASF integrability hypothesis, the elementary Fourier identity $\int \hat{u}(\xi)\hat{v}(\xi) d\xi = \int u(t)v(-t) dt$ when $u, v \in L^1$ and $\hat{u}\hat{v} \in L^1$, and the standard convolution inclusion $L^q * L^p \subset C_0$.

Novelty Check

The run indexes were searched for arXiv:2201.04421 and the keywords ‘abc’, ‘pqr’, ‘approximate Schauder frame’, ‘continuous approximate Schauder frame’, and ‘Gabor’. Adjacent frame packets were found, but no packet for this paper. Web searches on 2026-06-28 for exact phrases including ‘abc,pqr approximate Schauder frames’, ‘continuous approximate Schauder frame Gabor Lp’, and ‘tilde E tilde T approximate Schauder frame Lp’ found the source arXiv page and the Eisner–Freeman continuous Schauder frame paper, but no later answer to this same-sign continuous formulation.

References

1. K. Mahesh Krishna, *The (abc, pqr) -problem for Approximate Schauder Frames for Banach Spaces*, arXiv:2201.04421, <https://arxiv.org/abs/2201.04421>.
2. K. Mahesh Krishna, *Discrete and continuous Welch bounds for Banach spaces with applications*, arXiv:2201.00980.
3. Joseph Eisner and Daniel Freeman, *Continuous Schauder frames for Banach spaces*, arXiv:1812.08360.

Human Review Recommendation

Review as a likely valid counterexample/negative solution to the continuous problem in arXiv:2201.04421. The key convention to verify is that the source paper's dual pairing is bilinear, $[x, \omega] = \int x\omega$, and its dual modulation $\tilde{E}_b$ has the same sign as E_b .